Pipeline stages: What, Why, Who, and How (task sheet Exercise 1 Step 2)

Stage	What (inputs/output)	Why	Who	How (internal steps/tools)
Data ingestion	**Input:** new .pgn.zst **Output:** bronze objects + job metadata	Source of truth late binding to schema catch drift early	Data Engineer	1. Apache Airflow 2. Checksum validation 3. zstd decompression 4. Spark parser 5. Land to Minio bucket
Data pre-processing	**Input:** raw pgn **Output:** Delta tables, QA report	ML & SQL need clean types testable contracts	Data Engineer, ML Engineer	1. PySpark 2. Great Expectations 3. MinIO 4. Feast
Model development	**Input:** versioned feature slice **Output:** model + metrics in MLflow	Reproducible training avoid skew-serve mismatch	ML Engineer. Data Scientist	1. Temporal train/val/test split 2. LightGBM/XGBoost 3. MLflow (tracking + registry) 4. SHAP explainability 5. Registry gate (metric threshold)
Model deployment	**Input:** registered model **Output:** prediction service / batch	Deliver business value controlled rollout	ML Engineer MLOps	1. FastAPI 2. Blue-green deployment 3. shadow traffic / canary 4. Automatic rollback
Model monitoring	**Input:** live/batch scores + labels **Output:** alerts, retrain jobs	Drift and Data debt are ML-specific fallout models	SRE, ML engineer, data scientist on-call	1. Evidently (drift, data quality) 2. PSI / KS for drift detection 3. Grafana, Prometheus 4. Airflow retrain DAG